

STA442 Assignment 1

Due Date: October 2nd, 2026, 11:59pm

Instructions:

- **Any question that is labelled with [H] requires a handwritten solution.** This can be written on a piece of paper and scanned, or be written digitally on an iPad or tablet. **If you ignore these instructions, you will be given a grade of 0 for that question.**
- For questions without the label [H], they will require a mix of R-coding and interpretation. RMarkdown is recommended but not required for submission. Include necessary codes and outputs only. If your code is complicated, make sure you comment on essential implementations.
- **Please keep your response concise.** Most of the questions can be answered within at most 3-4 sentences. If you are writing an “essay” to answer a question, it is highly likely that some of them are incorrect, which would lead to point deduction.
- Please ensure responses are **legible, and neatly organized.**
- **Compile everything into one pdf and submit through Crowdmark.**

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1. [H] (5 marks) Read Watson et al. (2015), “Mild hypohydration increases the frequency of driver errors during a prolonged, monotonous driving task.” The study received considerable media attention, with findings sometimes summarized by headlines suggesting that *driving while dehydrated may be as dangerous as driving while intoxicated*.

Answer the following questions.

- (a) (1 marks) Identify the population and describe the sample.
- (b) (1 marks) Briefly describe the study design (2 – 3 sentences).
- (c) (1 marks) Consider the statement:

“Driving while dehydrated is just as dangerous as driving while drunk.”

Does the study itself provide sufficient evidence to support this statement? Explain.

- (d) (1 mark) Rewrite the statement in part (c) so that it accurately communicates the main finding of the study to a general audience without overstating the evidence. In 1–2 sentences, explain why your version is more appropriate.
 - (e) (1 marks) Briefly describe (1 – 2 sentences) one potential consequence of media reports overstating the findings of this study.
2. (5 marks) Five observations on y are to be taken when $x = 2, 6, 10, 14, 18$ (fixed). The true data generating process is

$$y_i = 15 + 3x_i + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 4^2). \quad (1)$$

- (a) (0.5 marks) Can we rewrite Equation (1) as follows? Explain briefly.

$$y_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(15 + 3x_i, 4^2).$$

- (b) **(1 marks)** Generate five normal random numbers with mean 0 and variance 4² in R using `set.seed(442)`. Consider these random numbers as the error terms ϵ_i for the five y observations at $x = 2, 6, 10, 14$, and 18 , and generate $y_1, \dots, y_5$.
Fit a linear regression to obtain $\hat{\beta}_0$ and $\hat{\beta}_1$ by fitting a straight line to the five observations. Also calculate $\hat{E}[y_{\text{new}}]$ when $x_{\text{new}} = 12$, and obtain a 95% confidence interval for $E[y_{\text{new}}]$ when $x_{\text{new}} = 12$.
- (c) **(2.5 marks)** Repeat part (b) 10,000 times, generating new random error terms each time, to answer the following questions:
- (i) Make a histogram of the 10,000 estimates of $\hat{\beta}_1$, and calculate the mean and standard deviation of the 10,000 estimates. Are your results consistent with theoretical expectations?
- Note:** Theoretically, in this example, $\text{Var}(\hat{\beta}_1) = 0.00625\sigma^2$, where σ^2 is the variance of the error terms.
- (ii) What proportion of the 10,000 95% confidence intervals for $E[y_{\text{new}}]$ when $x_{\text{new}} = 12$ contain the true value of $E[y_{\text{new}}]$? Is this result close to what you would theoretically expect? Explain.
- (d) **(1 mark)** Suppose you are presenting the confidence interval obtained in part (b) to someone with no background in statistics. In 2–3 sentences, explain what the 95% confidence interval means in plain language, including a brief explanation about why estimates have uncertainty. .

Source: Adapted from *Applied Linear Statistical Models* by Kutner et al.

3. **[H] (5 marks)** This question has two parts where you are asked to critique various practices of statistics.
- (a) For each of the following, determine whether it is an appropriate practice of statistics. If it is not, provide 1–2 sentences explaining why it is inappropriate.
 - (i) **(1 mark)** Before collecting any data, I specified a significance level of $\alpha = 0.05$. After conducting the study, I obtained a p -value of 0.08. Since the result was close to 0.05, I decided to use $\alpha = 0.10$ instead and concluded that the result was statistically significant.
 - (ii) **(1 mark)** I collected data on 20 potential predictors of an outcome. I examined the correlations between each predictor and the outcome and selected the predictor with the largest correlation. I then fit a simple linear regression using that predictor and reported the usual p -value from the regression as evidence of an association.
 - (iii) **(1 mark)** A researcher is interested in whether any of 50 different genes are associated with a disease outcome. The researcher conducts 50 hypothesis tests, each using $\alpha = 0.05$, and reports all p -values. The researcher also applies a Bonferroni correction when determining which associations are statistically significant.
 - (iv) **(1 mark)** After finding a statistically significant association between two variables, a researcher removed five observations they considered to be outliers and reanalyzed the data to assess whether the result was robust.
 - (b) **(1 mark)** There's a paper published in a Psychology journal, and one of the experiments used 2-way ANOVA and tested the interaction effect between experimental conditions A (2 factors) and B (4 factors). They used $n = 58$ and a p -value of 0.018 and reported a power of 75%. To present your new research, you replicated the designs with the same number of samples but you got a p -value of 0.21. A referee has asked the following question:

“Why is your result different from the results of the previous study? The previous study found statistical significance, but your study doesn’t.”

From the statistical perspective, how would you respond to the referee’s comment? Please assume that all research was carried out in accordance with research ethics. The knowledge of ANOVA is not needed as long as you know it is a statistical test. Please limit your response to 2 sentences.

4. (5 marks) In this problem, you will use R to simulate and analyze data from $m = 100$ fund managers. Use the code provided below to generate the data, and use R as needed to answer the questions.

```
set.seed(442)
n = 200
m = 100
X = matrix(rnorm(n*m), ncol = m)
```

These data represent each fund manager’s percentage returns for each of $n = 200$ months. We wish to test the null hypothesis that each fund manager’s percentage returns have population mean equal to zero. Notice that we simulated the data in such a way that each fund manager’s percentage returns do have population mean zero; in other words, all m null hypotheses are true.

- a (0.5 marks) If we control Type I error for each null hypothesis at level $\alpha = 0.05$, then how many null hypotheses do we reject in this data?
- b (1 mark) Assuming the 100 hypothesis tests are independent, what is the probability of making at least one Type I error if each hypothesis is tested at level $\alpha = 0.05$? Briefly interpret your answer.
- c (1 mark) What significance level, α , should be used for each individual hypothesis test in order to control the FWER at 0.05? Explain.
- d (0.5 marks) If we control the FWER at level 0.05 using (c), then how many null hypotheses do we reject?
- e (1 mark) Now suppose we “cherry-pick” the 10 fund managers who perform the best in our data, and the ‘performance’ is determined based on the average percentage returns. If we control the FWER for just these 10 fund managers at level 0.05, then how many null hypotheses do we reject in this data?
- f (1 marks) Explain briefly why the analysis in (e) is misleading.

Source: *An Introduction to Statistical Learning* by James et al.